

Unit II

Mechanism and Machines

Introduction

- Theory of Machines is a subject which deals with relative motion between parts of machines and forces acting between them.
- Theory of machines is sub-divided as; 1. Kinematics, 2. Dynamics, 3. Kinetics and 4. Statics
- 1. Kinematics- deals with the relative motion between the various parts of the machines,
- 2. Dynamics – is the study of the forces and their effects, while acting upon the machine parts in motion.
- 3. Kinetics – focuses on the inertia forces which arise from the combined effect of the mass and motion of the machine parts.
- 4. Statics- is the studies of the forces and their effects while the machine parts are at rest. The mass of the parts is assumed to be negligible.

Types of Motion

1. Rectilinear Motion

It is the simplest type of motion and is along a straight line path. Such a motion is also known as translatory motion.

2. Curvilinear Motion

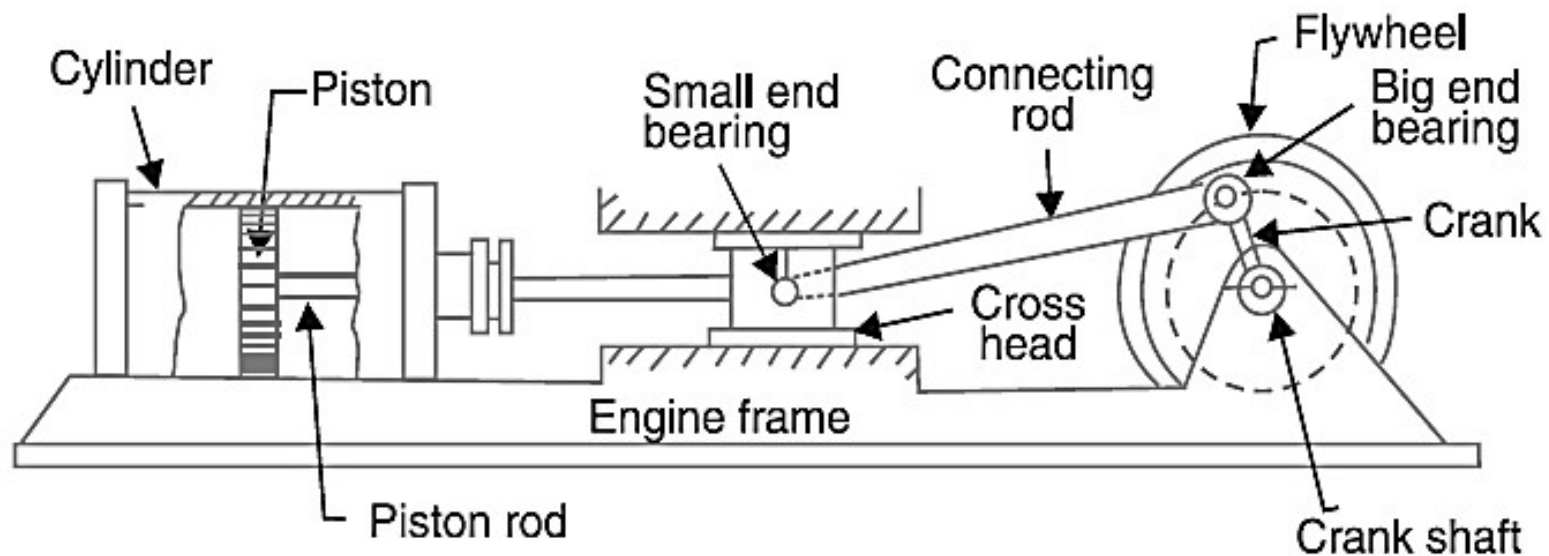
It is the motion along a curved path. Such a motion, when confined to one plane, is called plane curvilinear motion.

3. Planner Motion

When the motion of a body is confined to only one plane, the motion is said to be *plane motion*. The plane motion may be either rectilinear or curvilinear.

Kinematic Link or Element

- Each part of a machine, which moves relative to some other part, is known as a kinematic link (or simply link) or element.
- A link may consist of several parts, which are rigidly fastened together, so that they do not move relative to one another.



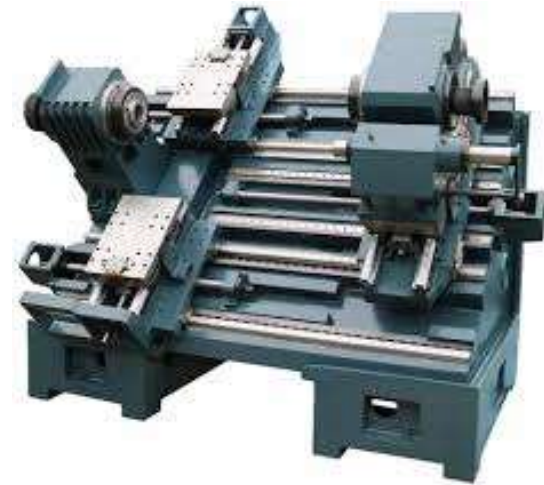
Types of Links

In order to transmit motion, the driver and the follower may be connected by the following three types of links :

- 1. Rigid link.** A rigid link is one which does not undergo any deformation while transmitting motion. Strictly speaking, rigid links do not exist. However, as the deformation of a connecting rod, crank etc. of a reciprocating steam engine is not appreciable, they can be considered as rigid links.
- 2. Flexible link.** A flexible link is one which is partly deformed in a manner not to affect the transmission of motion. For example, belts, ropes, chains and wires are flexible links and transmit tensile forces only.
- 3. Fluid link.** A fluid link is one which is formed by having a fluid in a receptacle and the motion is transmitted through the fluid by pressure or compression only, as in the case of hydraulic presses, jacks and brakes.

Structure

It is an assemblage of a number of resistant bodies (known as members) having no relative motion between them and meant for carrying loads having straining action. A railway bridge, a roof truss, machine frames etc., are the examples of a structure.



Difference Between a Machine and a Structure

The following differences between a machine and a structure are important from the subject point of view :

1. The parts of a machine move relative to one another, whereas the members of a structure do not move relative to one another.
2. A machine transforms the available energy into some useful work, whereas in a structure no energy is transformed into useful work.
3. The links of a machine may transmit both power and motion, while the members of a structure transmit forces only.

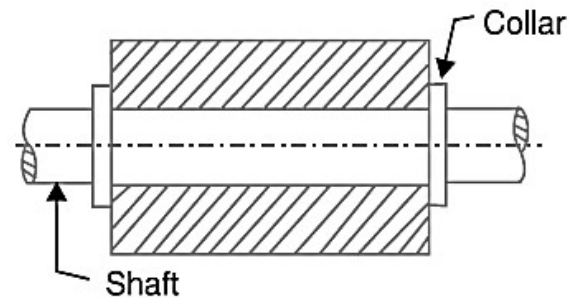
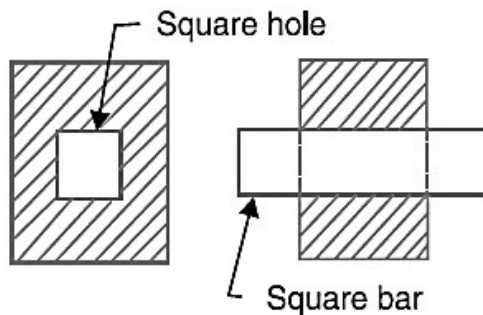
Kinematic Pair

The two links or elements of a machine, when in contact with each other, are said to form a pair. If the relative motion between them is completely or successfully constrained (i.e. in a definite direction), the pair is known as kinematic pair.

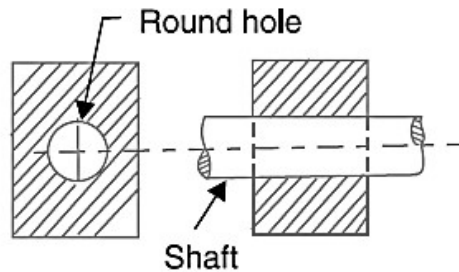
Types of Constrained Motions

Following are the three types of constrained motions :

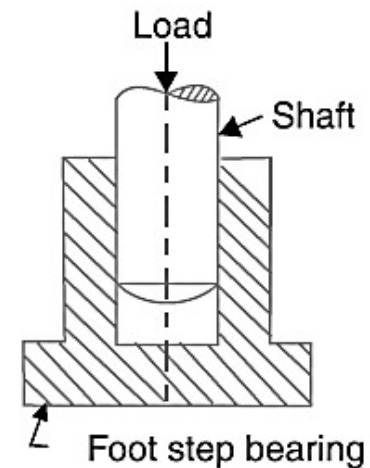
1. Completely constrained motion. When the motion between a pair is limited to a definite direction irrespective of the direction of force applied, then the motion is said to be a completely constrained motion.



2. Incompletely constrained motion. When the motion between a pair can take place in more than one direction, then the motion is called an incompletely constrained motion.



3. Successfully constrained motion. When the motion between the elements, forming a pair, is such that the constrained motion is not completed by itself, but by some other means, then the motion is said to be successfully constrained motion.



Classification of Kinematic Pairs

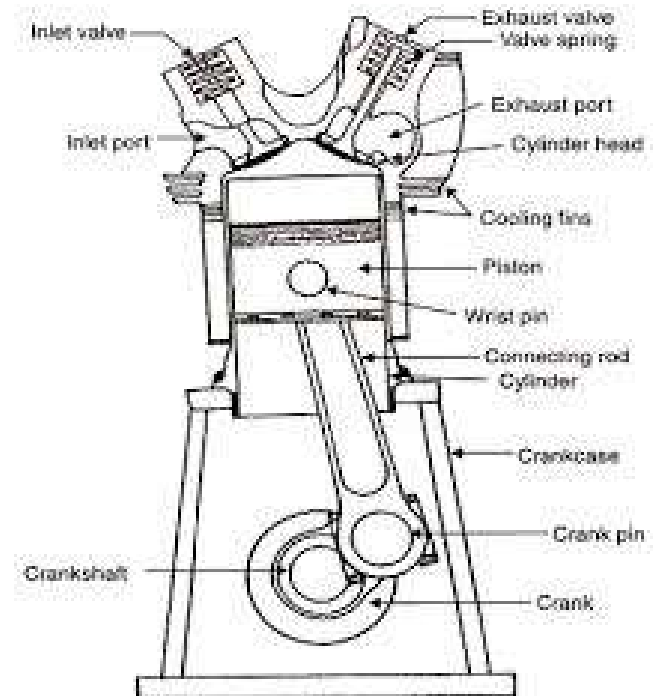
The kinematic pairs may be classified according to the following considerations :

1. According to the type of relative motion between the elements. The kinematic pairs according to type of relative motion between the elements may be classified as discussed below:

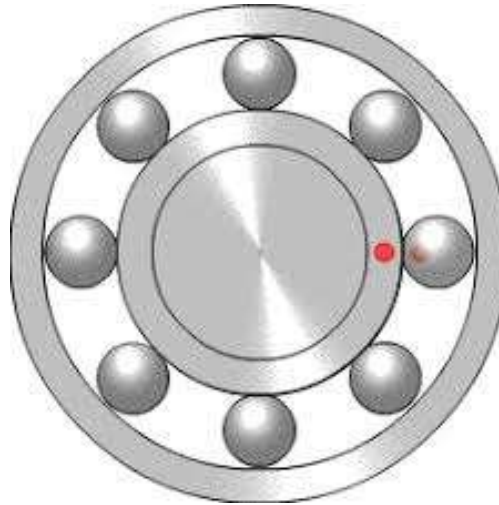
(a) Sliding pair. When the two elements of a pair are connected in such a way that one can only slide relative to the other, the pair is known as a sliding pair. The piston and cylinder, cross-head and guides of a reciprocating steam engine, ram and its guides in shaper, tail stock on the lathe bed etc. are the examples of a sliding pair.



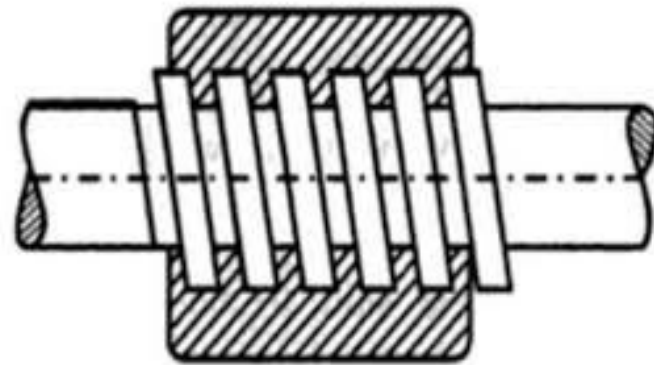
(b) Turning pair. When the two elements of a pair are connected in such a way that one can only turn or revolve about a fixed axis of another link, the pair is known as turning pair. A shaft with collars at both ends fitted into a circular hole, the crankshaft in a journal bearing in an engine, lathe spindle supported in head stock, cycle wheels turning over their axles etc. are the examples of a turning pair. A turning pair also has a completely constrained motion.



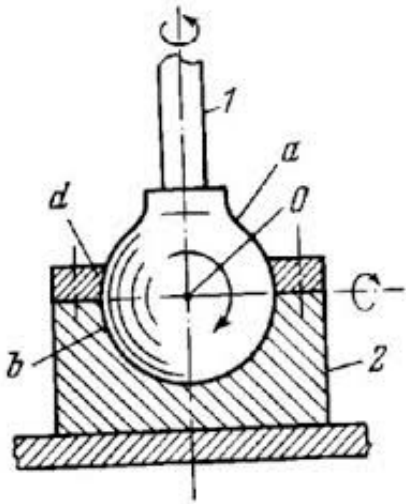
(c) Rolling pair. When the two elements of a pair are connected in such a way that one rolls over another fixed link, the pair is known as rolling pair. Ball and roller bearings are examples of rolling pair.



(d) Screw pair. When the two elements of a pair are connected in such a way that one element can turn about the other by screw threads, the pair is known as screw pair. The lead screw of a lathe with nut, and bolt with a nut are examples of a screw pair.

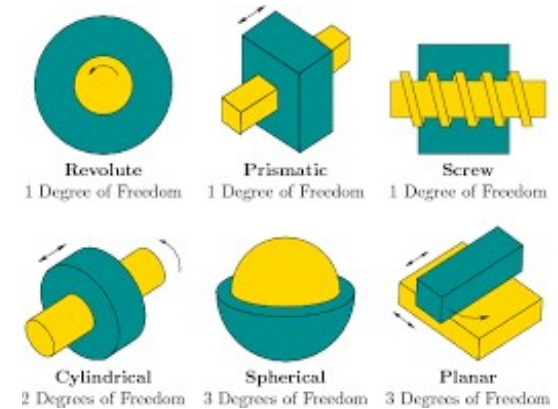


(e) Spherical pair. When the two elements of a pair are connected in such a way that one element (with spherical shape) turns or swivels about the other fixed element, the pair formed is called a spherical pair. The ball and socket joint, attachment of a car mirror, pen stand etc.. are the examples of a spherical pair.



2. According to the type of contact between the elements. The kinematic pairs according to the type of contact between the elements may be classified as discussed below :

(a) Lower pair. When the two elements of a pair have a surface contact when relative motion takes place and the surface of one element slides over the surface of the other, the pair formed is known as lower pair. It will be seen that sliding pairs, turning pairs and screw pairs form lower pairs.



(b) Higher pair. When the two elements of a pair have a line or point contact when relative motion takes place and the motion between the two elements is partly turning and partly sliding, then the pair is known as higher pair. A pair of friction discs, toothed gearing, roller bearings and cam and follower are the examples of higher pairs.

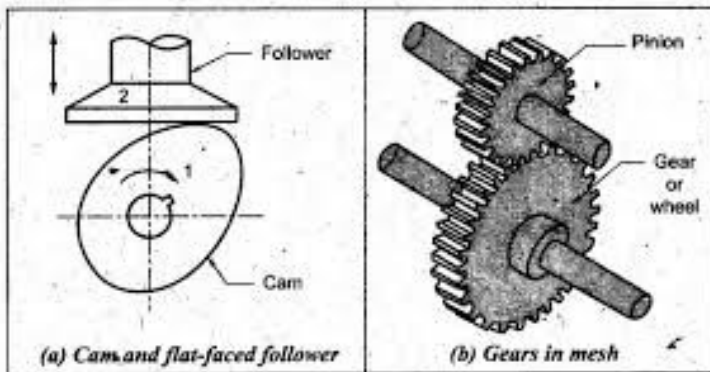
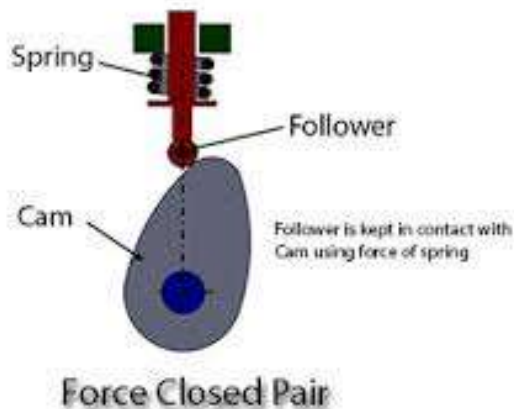
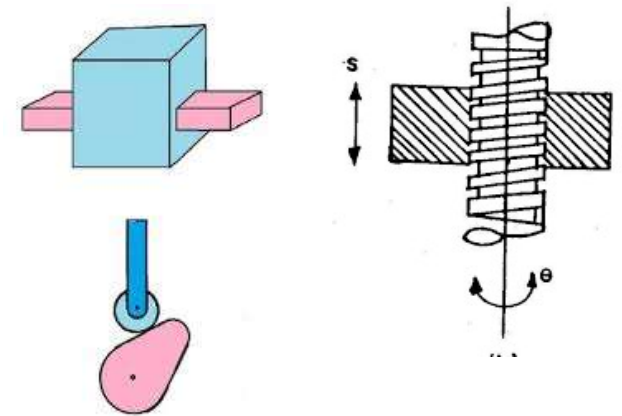


Fig. 1.17. Higher pairs

3. According to the type of closure. The kinematic pairs according to the type of closure between the elements may be classified as discussed below :

(a) Self closed pair. When the two elements of a pair are connected together mechanically in such a way that only required kind of relative motion occurs, it is then known as self closed pair. The lower pairs are self closed pair.



(b) Force - closed pair. When the two elements of a pair are not connected mechanically but are kept in contact by the action of external forces, the pair is said to be a force-closed pair. The cam and follower is an example of force closed pair, as it is kept in contact by the forces exerted by spring and gravity.

Kinematic Chain

When the kinematic pairs are coupled in such a way that the last link is joined to the first link to transmit definite motion (i.e. completely or successfully constrained motion), it is called a kinematic chain.

If each link is assumed to form two pairs with two adjacent links, then the relation between the number of pairs (p) forming a kinematic chain and the number of links (l) may be expressed in the form of an equation :

$$l = 2p - 4$$

Another relation between the number of links (l) and the number of joints (j) which constitute a kinematic chain is given by the expression :

$$j = \frac{3}{2} l - 2$$

Consider the arrangement of three links AB , BC and CA with pin joints at A , B and C as shown in Fig. In this case,

Number of links, $l = 3$

Number of pairs, $p = 3$

and number of joints, $j = 3$

From equation $l = 2p - 4$

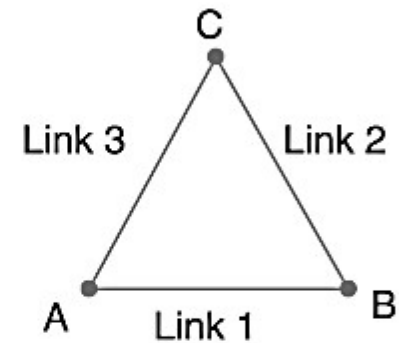
or $3 = 2 \times 3 - 4 = 2$

i.e. L.H.S. > R.H.S.

Now from equation

$$j = \frac{3}{2} l - 2 \quad \text{or} \quad 3 = \frac{3}{2} \times 3 - 2 = 2.5$$

L.H.S. > R.H.S.



Since both the conditions are satisfied, therefore this chain is a locked chain and forms a rigid frame

Consider the arrangement of four links AB , BC , CD and DA as shown in Fig. In this case

$$l = 4, p = 4, \text{ and } j = 4$$

From equation

$$l = 2p - 4$$

$$4 = 2 \times 4 - 4 = 4$$

i.e.

$$\text{L.H.S.} = \text{R.H.S.}$$

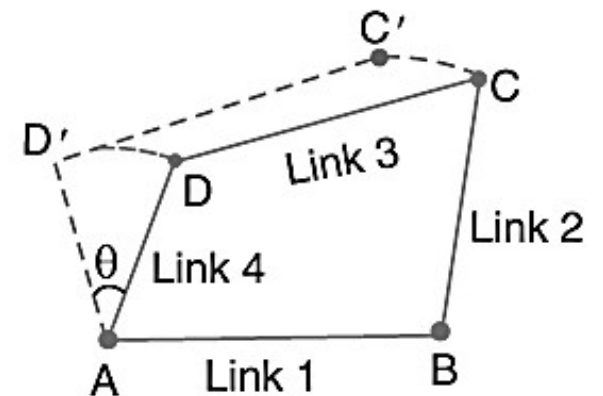
From equation

$$j = \frac{3}{2}l - 2$$

$$4 = \frac{3}{2} \times 4 - 2 = 4$$

i.e.

$$\text{L.H.S.} = \text{R.H.S.}$$

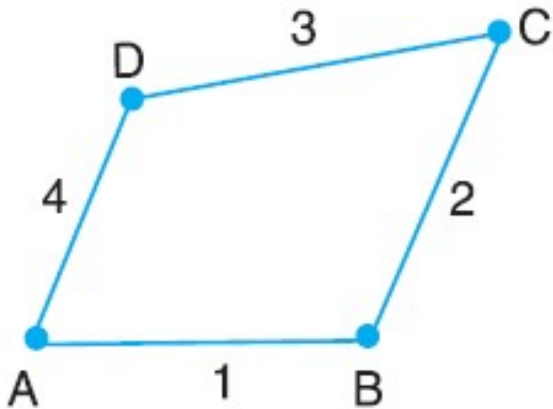


In this case both conditions are satisfied therefore this chain is a kinematic chain

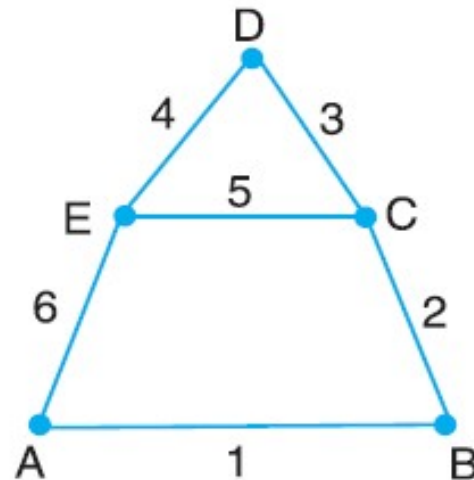
Types of Joints in a Chain

Binary Chain –

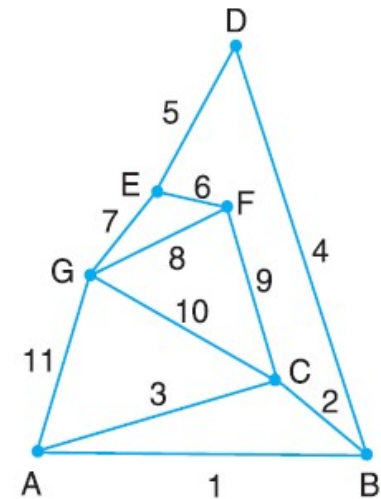
When two links are joining at a same connection



Ternary joint - When three links are joined at the same connection, the joint is known as ternary joint.



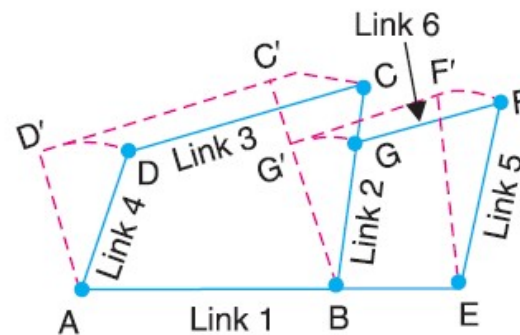
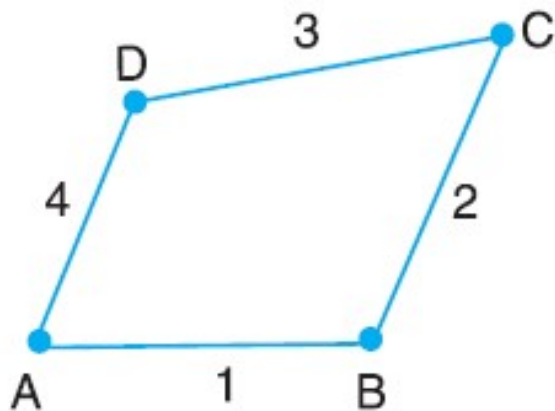
Quaternary joint- When four links are joined at the same connection, the joint is called a quaternary joint.



Mechanism

When one of the links of a kinematic chain is fixed, the chain is known as *mechanism*. It may be used for transmitting or transforming motion e.g. engine indicators, typewriter etc.

- A mechanism with four links is known as *simple mechanism*,
- A mechanism with more than four links is known as *compound mechanism*.
- When a mechanism is required to transmit power or to do some particular type of work, it then becomes a machine.

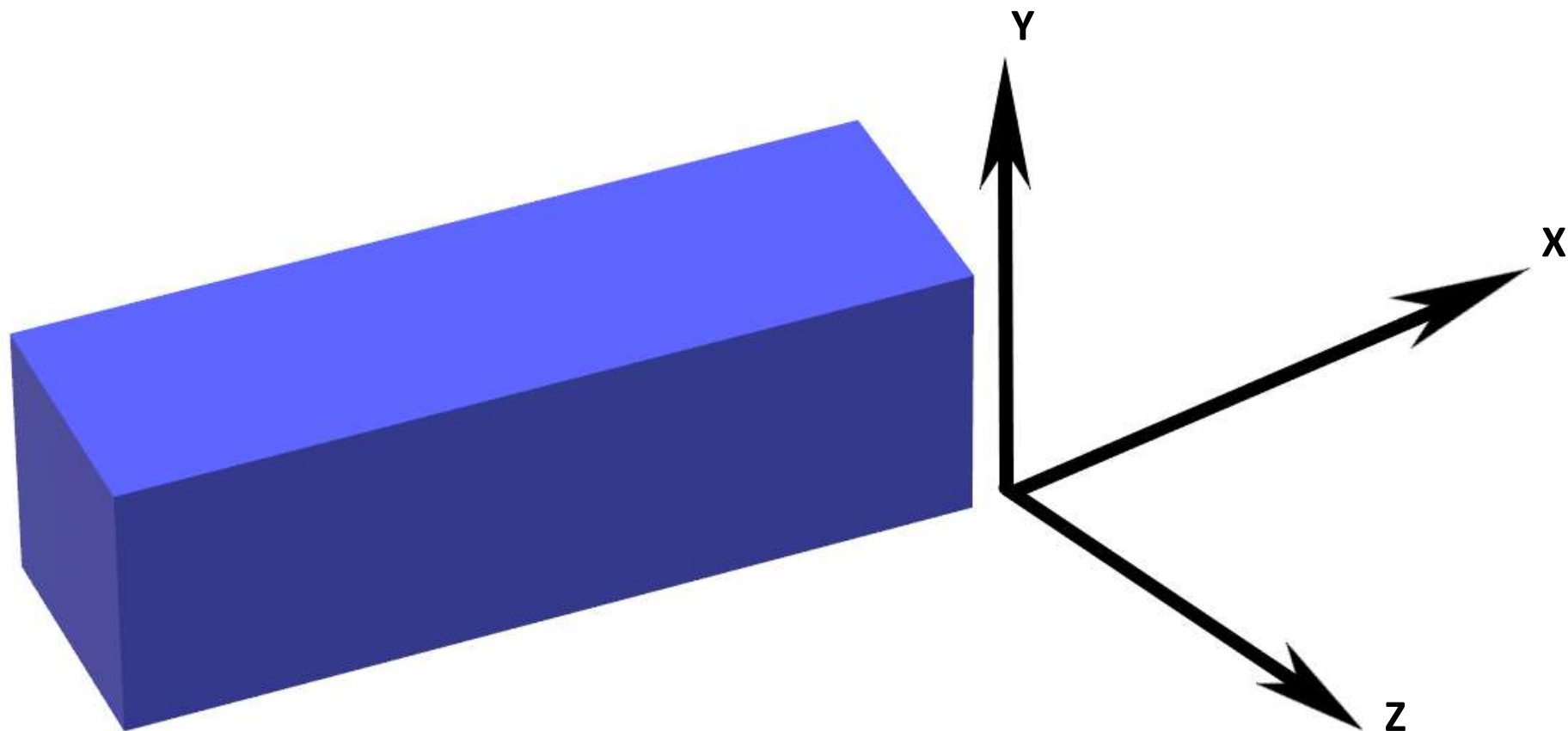


Number of Degrees of Freedom for Plane Mechanisms

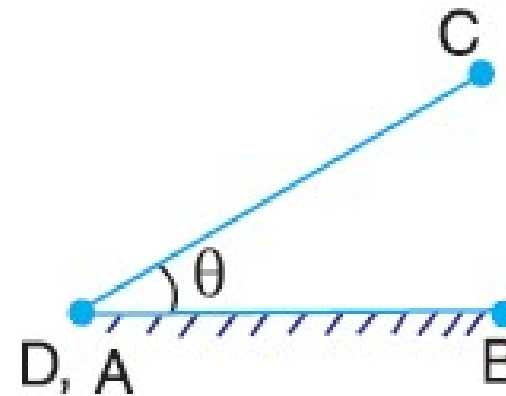
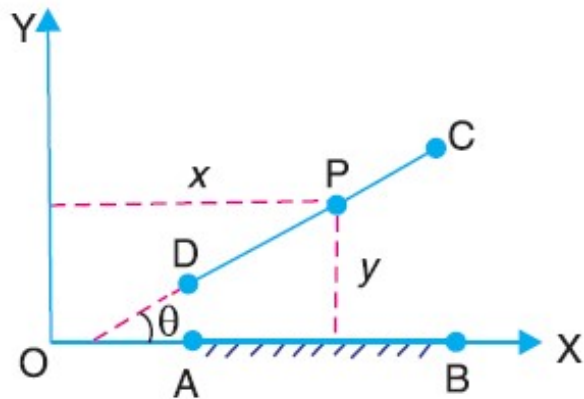
In the design or analysis of a mechanism, one of the most important concern is the number of degrees of freedom (also called movability) of the mechanism.

It is defined as the number of input parameters (usually pair variables) which must be independently controlled in order to bring the mechanism into a useful engineering purpose.

In simple words degrees of freedom is number of ways a body can move in space. The maximum possible degrees of freedom for a link in space is six. There are three linear movement along the axis called axial movement and three rotational movement about the axis.



Identifying degree of freedom of binary Joint in planer Motion



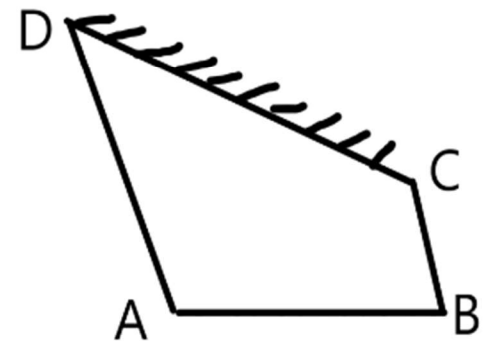
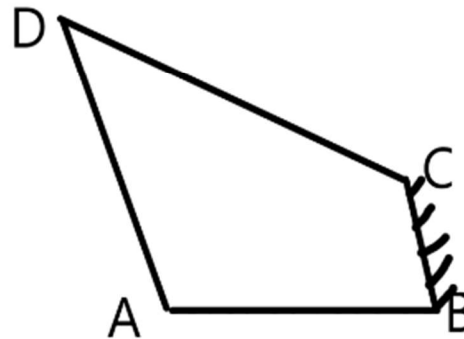
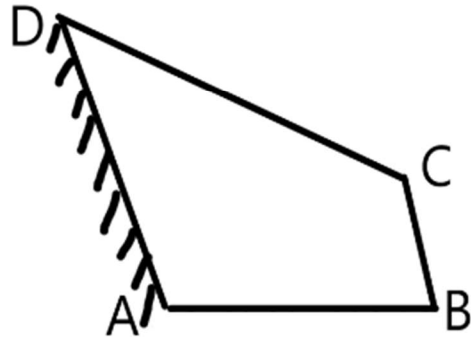
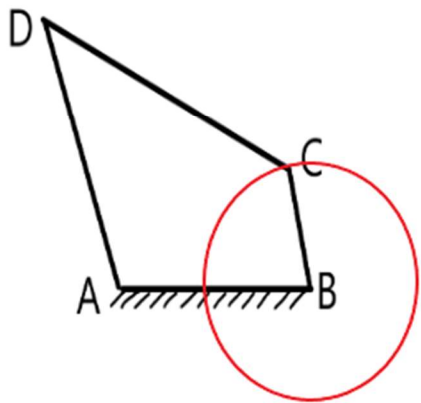
Degrees of freedom n can be determined as;

$$n = 3(l-1) - 2j - h$$

Where n = degrees of freedom, l = number of links, j = number of joints and h = number of higher pairs

Inversion of Mechanism

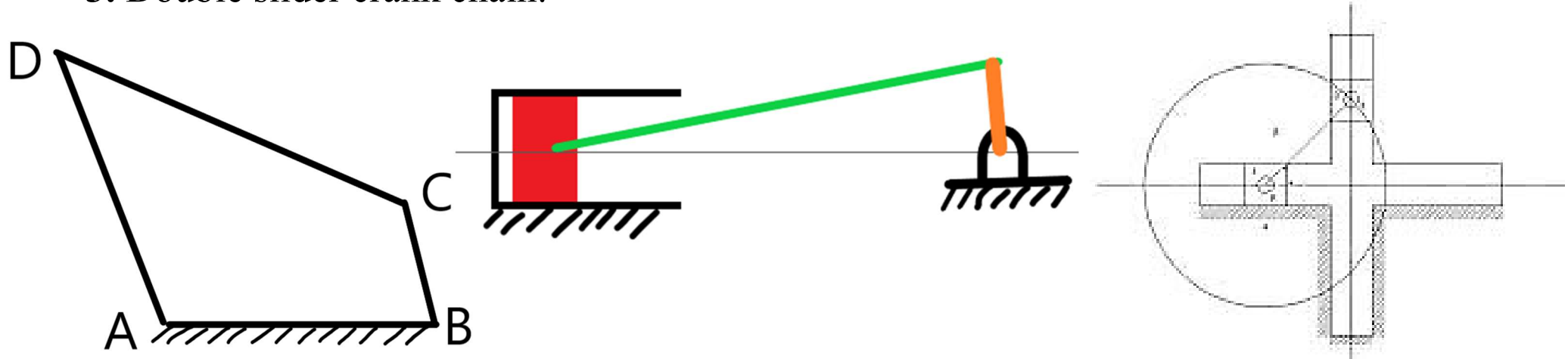
- When one of links is fixed in a kinematic chain, it is called a mechanism.
- Many mechanisms can be obtained as the number of links in a kinematic chain by fixing, in turn, different links in a kinematic chain.
- This method of obtaining different mechanisms by fixing different links in a kinematic chain, is known as *inversion of the mechanism*.



Types of Kinematic Chains

The most important kinematic chains are those which consist of four lower pairs, each pair being a sliding pair or a turning pair. The following three types of kinematic chains with four lower pairs are important :

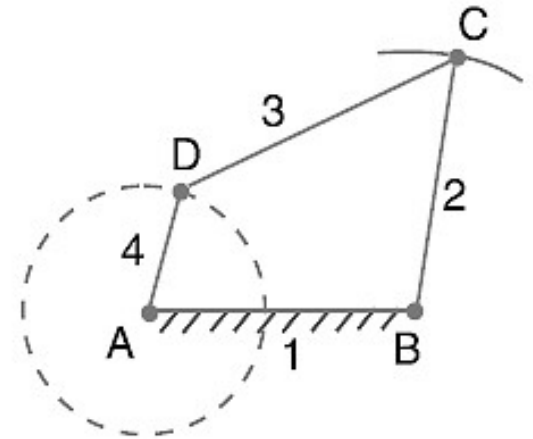
1. Four bar chain or quadric cyclic chain,
2. Single slider crank chain, and
3. Double slider crank chain.



Four Bar Chain or Quadric Cycle Chain

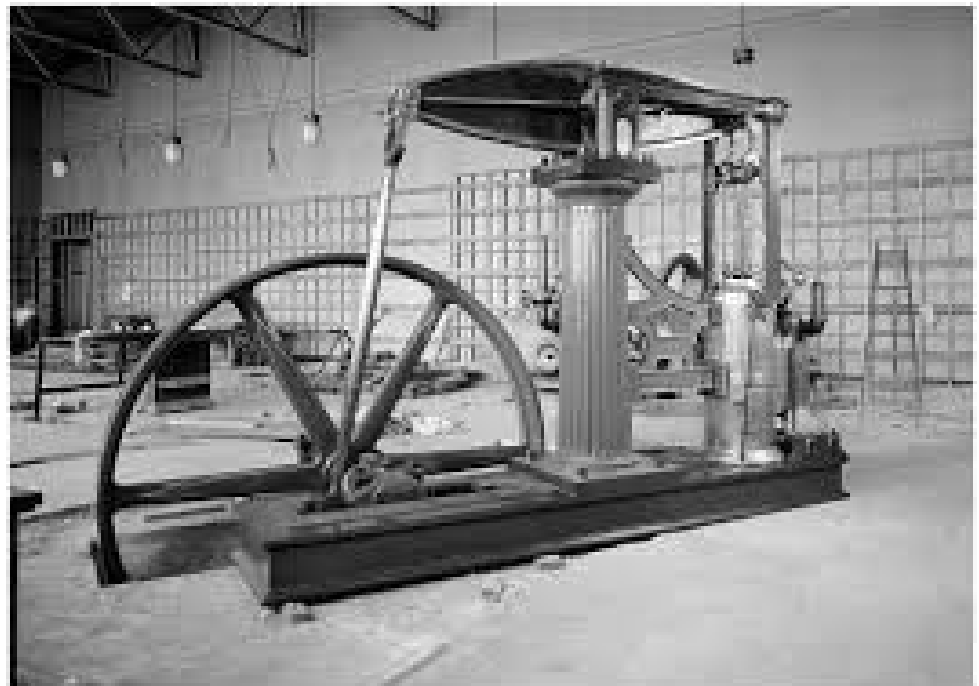
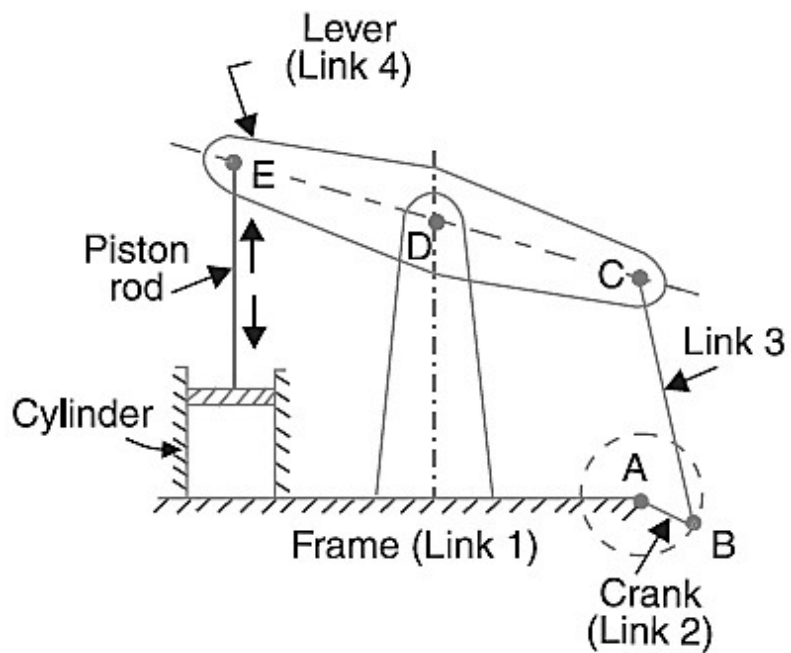
- The four links may be of different lengths.
- According to **Grashof's law** for a four bar mechanism, the sum of the shortest and longest link lengths should not be greater than the sum of the remaining two link lengths if there is to be continuous relative motion between the two links.

$$AD + CD \leq AB + BC$$

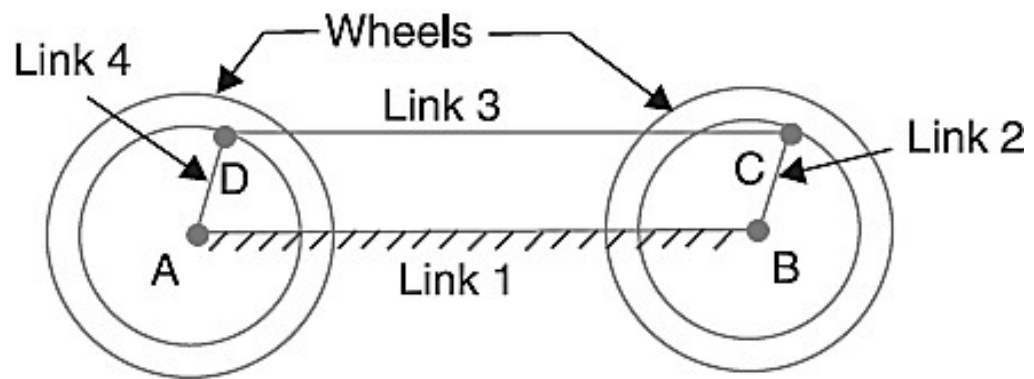


Inversion of four bar chain

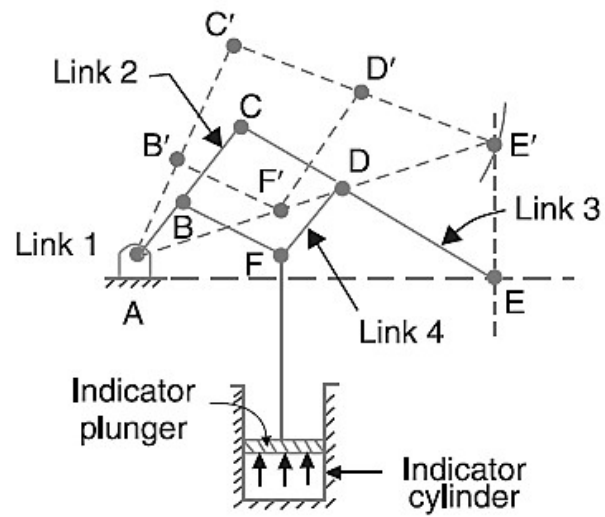
Beam engine (crank and lever mechanism)



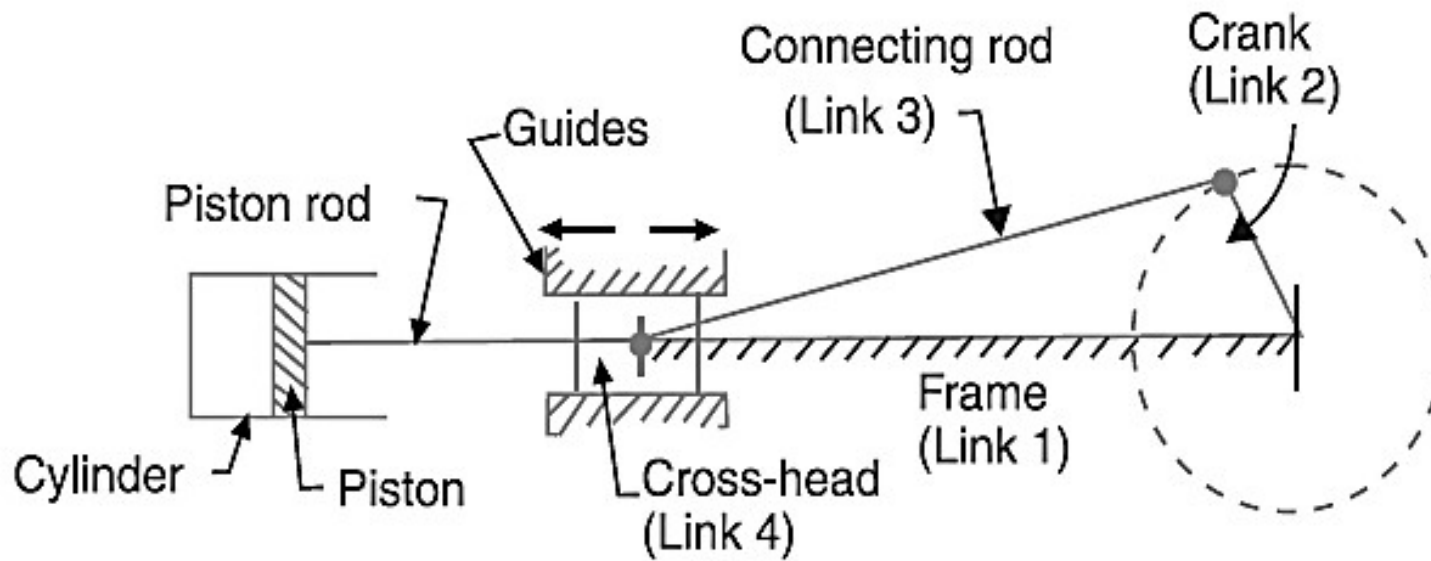
Coupling rod of a locomotive (Double crank mechanism)



Watt's indicator mechanism (Double lever mechanism)

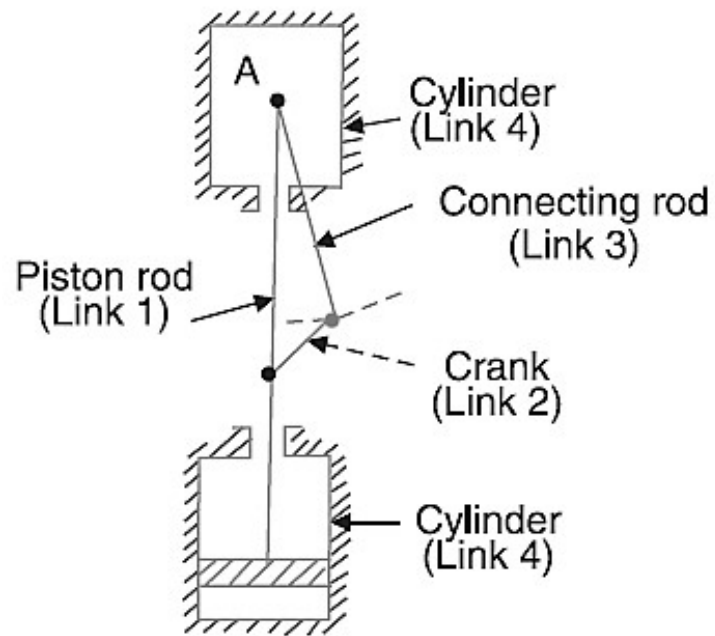


Single Slider Crank Chain

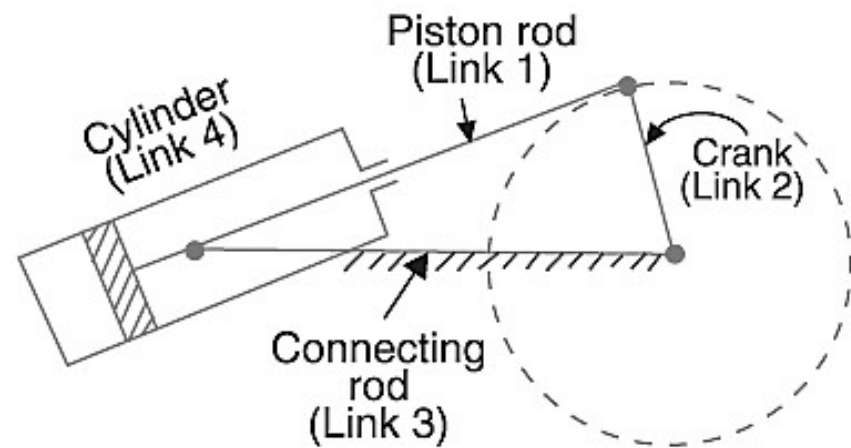


Inversions of Single Slider Crank Chain

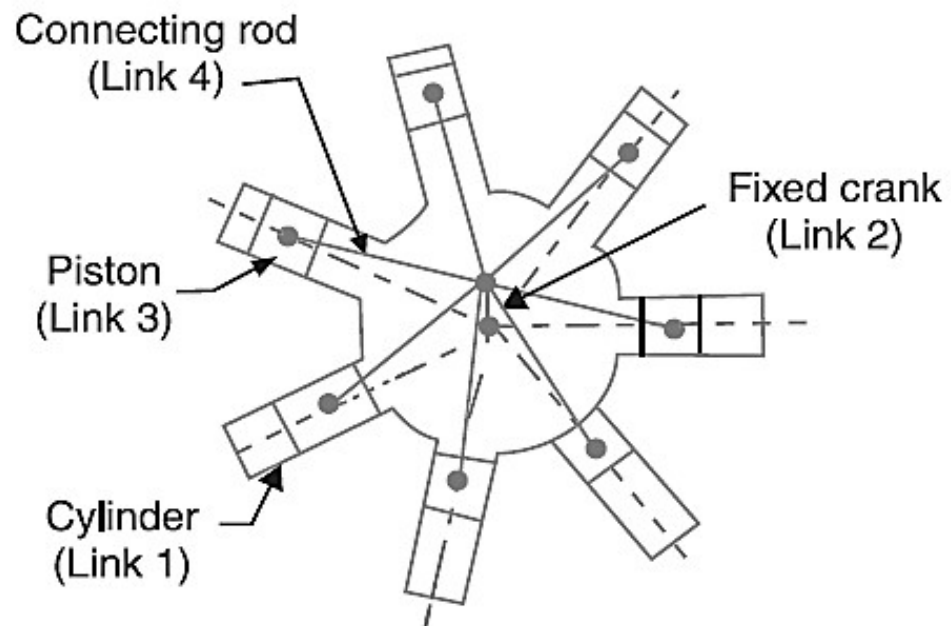
Pendulum pump or Bull engine



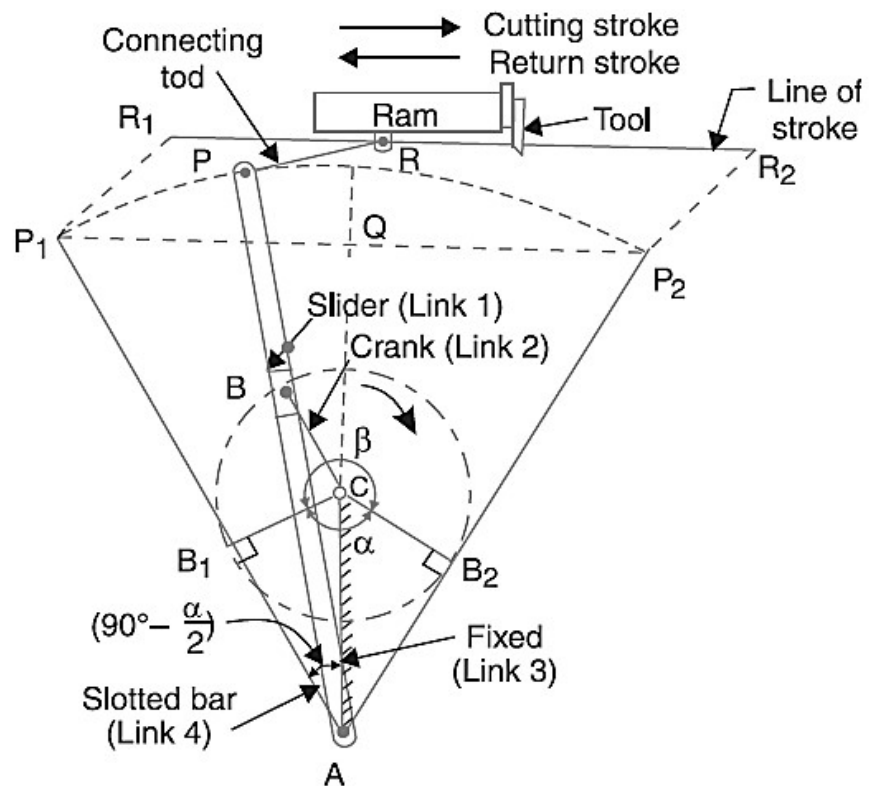
Oscillating cylinder engine



Rotary internal combustion engine or Gnome engine



Crank and slotted lever quick return motion mechanism



Ratio of cutting stroke to return stroke and stroke length

Referring to the figure in the previous slide

$$\frac{\text{Time of cutting stroke}}{\text{Time of return stroke}} = \frac{\beta}{\alpha} = \frac{\beta}{360^\circ - \beta} \quad \text{or} \quad \frac{360^\circ - \alpha}{\alpha}$$

Since the tool travels a distance of $R_1 R_2$, during cutting and return stroke, therefore travel of the tool or length of stroke

$$\begin{aligned} &= R_1 R_2 = P_1 P_2 = 2P_1 Q = 2AP_1 \sin \angle P_1 A Q \\ &= 2AP_1 \sin \left(90^\circ - \frac{\alpha}{2} \right) = 2AP \cos \frac{\alpha}{2} \quad \dots (\because AP_1 = AP) \\ &= 2AP \times \frac{CB_1}{AC} \quad \dots \left(\because \cos \frac{\alpha}{2} = \frac{CB_1}{AC} \right) \\ &= 2AP \times \frac{CB}{AC} \quad \dots (\because CB_1 = CB) \end{aligned}$$

A crank and slotted lever mechanism used in a shaper has a center distance of 300 mm between the center of oscillation of the slotted lever and the center of rotation of the crank. The radius of the crank is 120 mm and length of slotted lever is 800 mm. Find the stroke length and ratio of the time of cutting to the time of return stroke. Also determine percentage of cutting and return time.

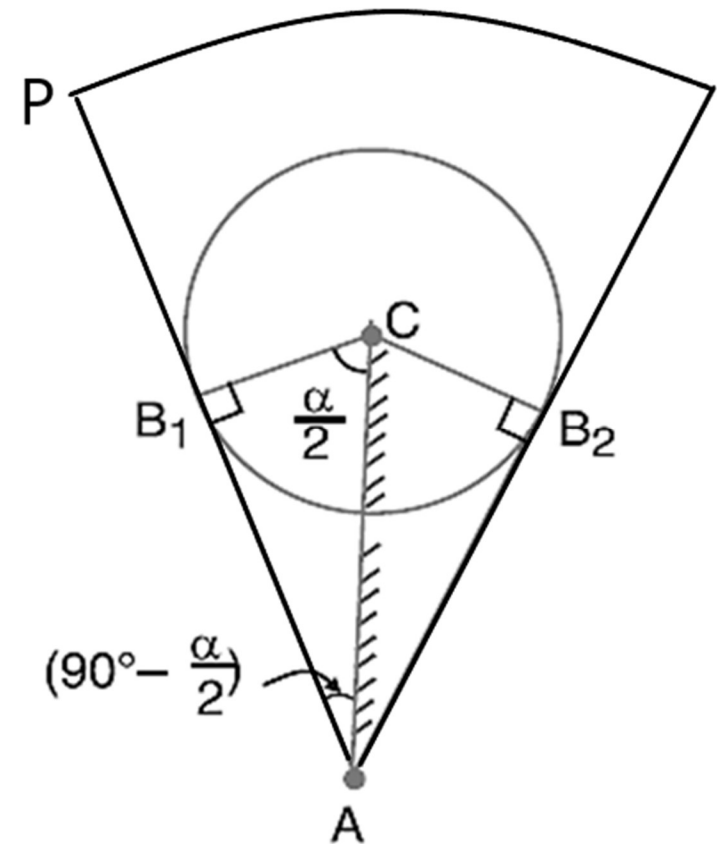
Given : $AC = 300$ mm ; $CB_1 = 120$ mm, $AP = 800$ mm

The extreme positions of the crank are shown in Fig.

We know that

1. Stroke Length

$$\begin{aligned}\text{Stroke Length} &= 2AP \times CB/AC \\ &= 2 \times 800 \times 120 / 300 \\ &= 640 \text{ mm}\end{aligned}$$



Cutting to return stroke time ratio

$$\sin \angle CAB_1 = \sin(90^\circ - \alpha/2)$$

$$= \frac{CB_1}{AC} = \frac{120}{300} = 0.4$$

$$\therefore \angle CAB_1 = 90^\circ - \alpha/2$$

$$= \sin^{-1} 0.4 = 23.6^\circ$$

$$\alpha/2 = 90^\circ - 23.6^\circ = 66.4^\circ$$

$$\alpha = 2 \times 66.4 = 132.8^\circ$$

$$\begin{aligned} \text{Angle in forward stroke } \beta &= 360 - \alpha \\ &= 360 - 132.8 \\ &= 227.2 \end{aligned}$$

We know that

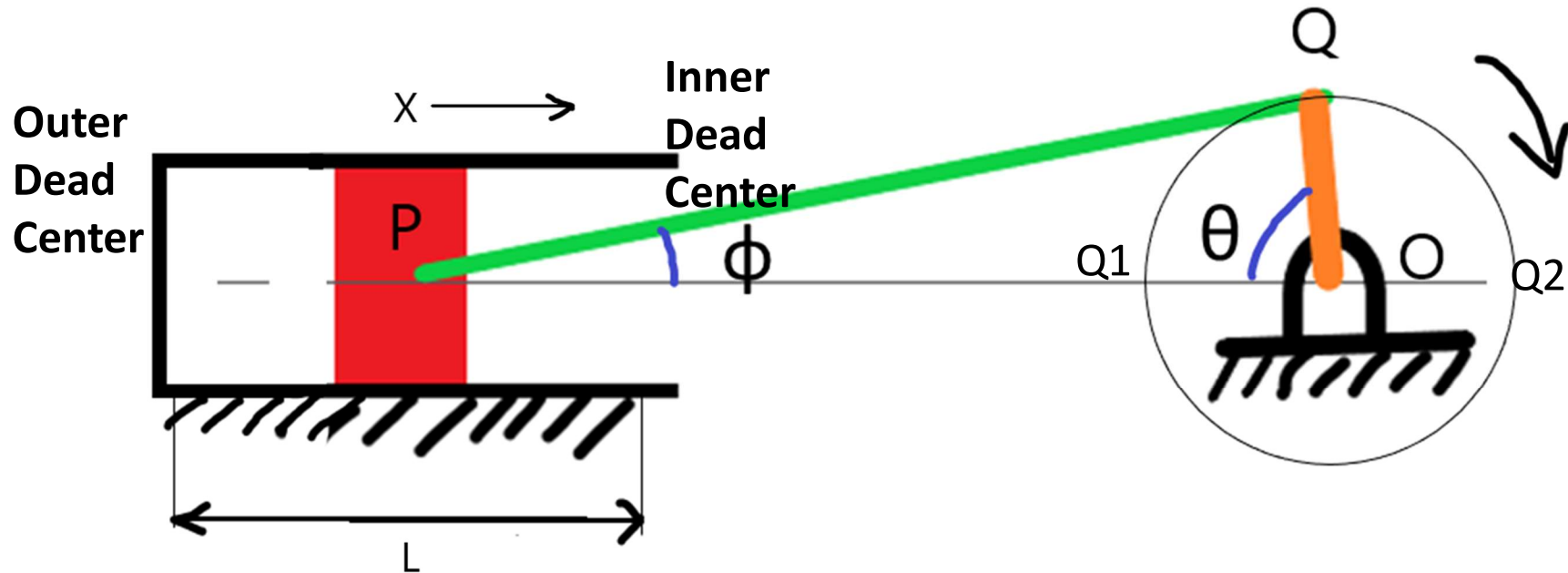
$$\frac{\text{Time of cutting stroke}}{\text{Time of return stroke}} = \frac{360^\circ - \alpha}{\alpha} = \frac{360^\circ - 132.8^\circ}{132.8^\circ} = 1.72$$

$$\% \text{ of cutting time} = \beta \times 100 / 360 = 227.2 \times 100 / 360 = 63.1 \%$$

$$\% \text{ of return time} = \alpha \times 100 / 360 = 132.8 \times 100 / 360 = 36.8 \%$$

00φ

Displacement of Slider in single Slider Crank Mechanism



Let L = Stroke Length, N = Rotational speed of crank in RPM

ϕ = Angle subtended by connecting rod with center line joining slider pin and crank shaft center

θ = Angle subtended by crank with center line

$PQ = l$ = length of connecting rod, $OQ = r$ radius of the crank, $n = l/r$

Displacement of the slider ODC to IDC = L

During the slider movement from ODC to IDC crank rotation = 180 deg

The point Q moves from Q1 to Q2 the distance $Q_1Q_2 = 2OQ_1 = 2r$

Therefor $L = 2r$

Displacement of the slider L = twice the crank length

Displacement of Slider

$$X = r \left[(1 - \cos \theta) + \frac{\sin^2 \theta}{2n} \right]$$

Distance travelled by the slider in one minute

Angle turned by the crank in X distance travelled by slider = π

Angle turned by the crank in 2X distance = 2π

Since the crank makes N rotations per minutes therefor the distance travelled by the slider per minute is = $2L N = 4rN$

Relation between l , r , ϕ , and θ

$$\frac{\sin \phi}{r} = \frac{\sin \theta}{l}$$

In single slider crank mechanism, length of the connecting rod is 160 mm and ratio of connecting rod to crank is 4. The crank rotates at 6000 RPM. Determine: 1. Stroke Length, 2. Distance travelled by the slider per second, 3. Angular displacement of connecting rod, 4. Displacement of slider when crank angle is 75 degree.

Given Data: Connecting Rod Length $l = 160$ mm, $n = l/r = 4$, $N = 6000$ RPM

Crank Radius $r = 160/4 = 40$ mm

1. Stroke Length $L = 2r = 2 \times 40 = 80$ mm

2. Distance travelled by the slider per second = $2LN/60 = 2 \times 80 \times 6000/60 = 16000$ mm or 16 m

3. Angular displacement of connecting rod ϕ varies from 0 to max. Maximum displacement will occur when crank angle $\theta = 90$ deg

$$\frac{\sin \phi}{r} = \frac{\sin \theta}{l}, \sin \phi = r \frac{\sin \theta}{l} = \frac{\sin \theta}{n} = \frac{\sin 90}{4} = 0.25, \phi = 14.47 \text{ degree}$$

4. Displacement of slider when crank angle is 75 degree

$$X = r \left[(1 - \cos \theta) + \frac{\sin^2 \theta}{2n} \right]$$

$$X = 40 \left[(1 - \cos 75) + \frac{\sin^2 75}{2 \times 4} \right]$$

$$X = 34.31 \text{ mm}$$

Power Transmission

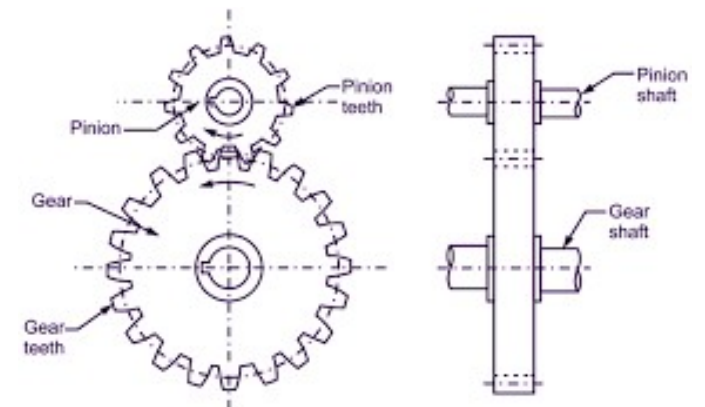
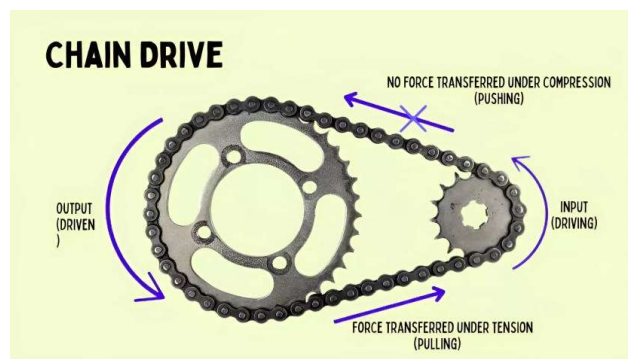
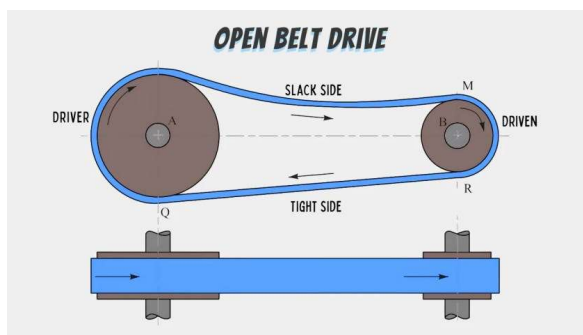
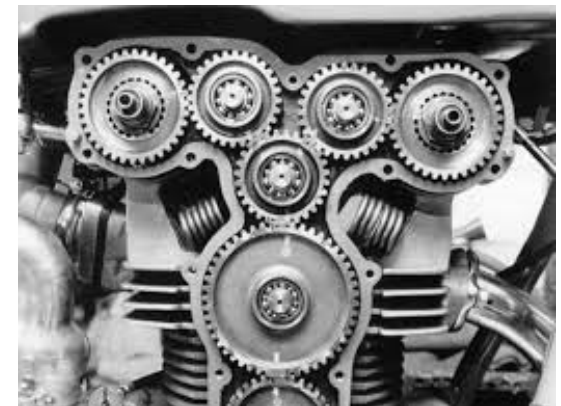
Belt drive



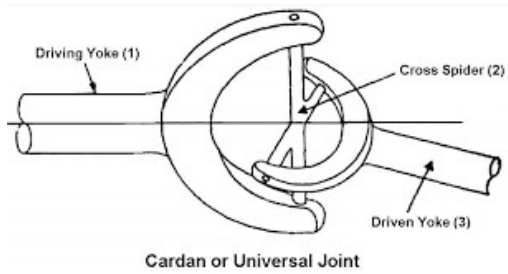
Chain drive



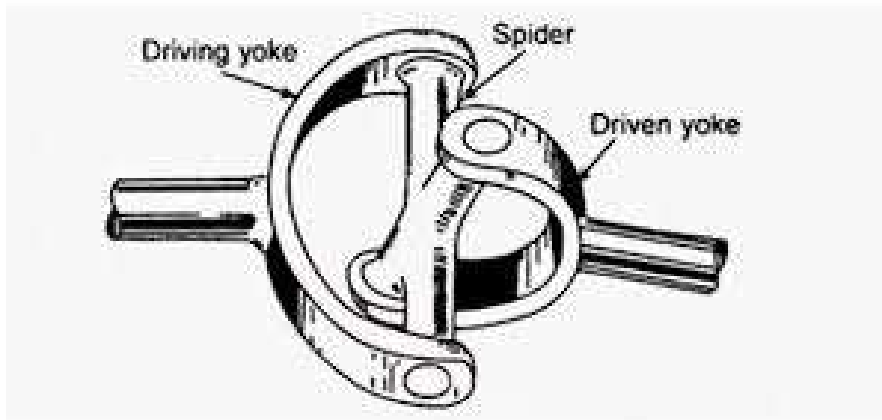
Gear drive



Hooke's Joint



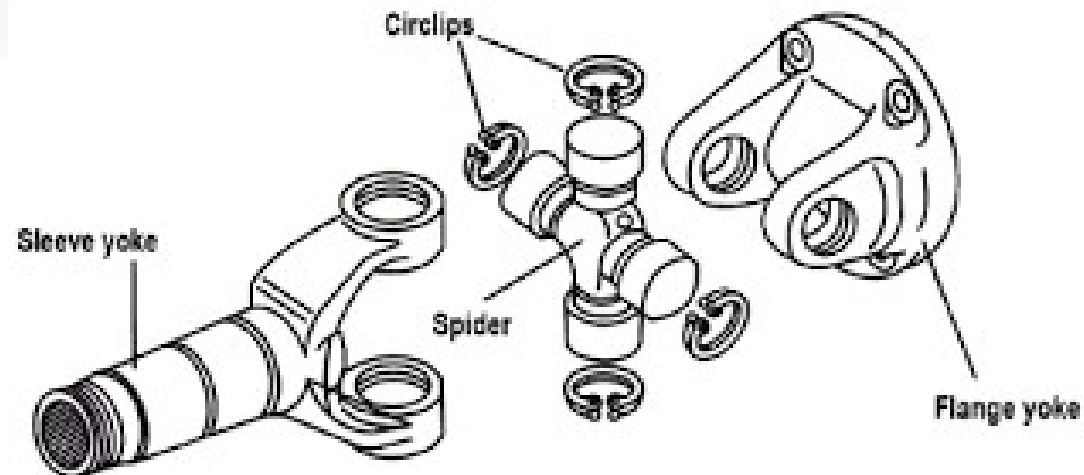
Parts of Hooke's Joint



Example

- Power from engine to front or rear wheel of the four wheeled vehicles

- It is used to transmit power when the driving and driven shaft are parallel but not coaxial
- It is also used power transmission when the shafts are not parallel and their axis are intersecting



Gears

- Gear drive is a positive and smooth drive, which transmits exact velocity ratio.
- **The gear is defined as a toothed element which is used for transmitting rotary motion from one shaft to another.**
- The distance between the two shafts is very small, then gears are used to transmit



Gear Classification

The following are the important classification of gears :

1. Classification based on the position of axes of the shafts.

The axes of the two shafts between which motion is to be transmitted may be

(a) Parallel, intersecting, and non-parallel and non-intersecting.

(a) Parallel Shafts - The following are the main types of gears to join parallel shafts :

(i) Spur gears

(ii) Helical gears

(iii) Double-helical gears.



(b) Intersecting Shaft

The gears used to connect two intersecting shafts are known as bevel gears. If the teeth on the gears are straight radial to the point of intersection of shaft axes then gears are known as straight bevel. But if teeth are inclined then gears are known as helical bevel (or spiral bevel) as shown in Figs.



(c) Non-parallel and Non-intersecting Shafts

The gears used to connect two non-parallel and non-intersecting shafts, are known as skew bevel gears or skew spiral gears.

TEL-SCIENCE.COM



2. Classification based on the peripheral velocity of the gears. According to the peripheral velocity, the gears are classified as :

- (a) Low velocity gears (2 m/s)
- (b) Medium velocity gears and (3 to 15 m/s)
- (c) High velocity gears (above 15 m/s)

3. Classification based on the position of teeth on the wheel. According to the position of the teeth on the wheel, the gears are classified as :

- (a) Straight – Spur Gear
- (b) Inclined – Helical Gears
- (c) Curved – Spiral Gears

4. Classification based on type of gearing.

(a) External Gearing



(b) Internal Gearing and

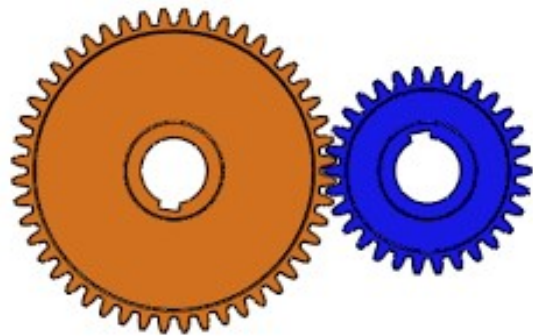


(c) Rack and Pinion

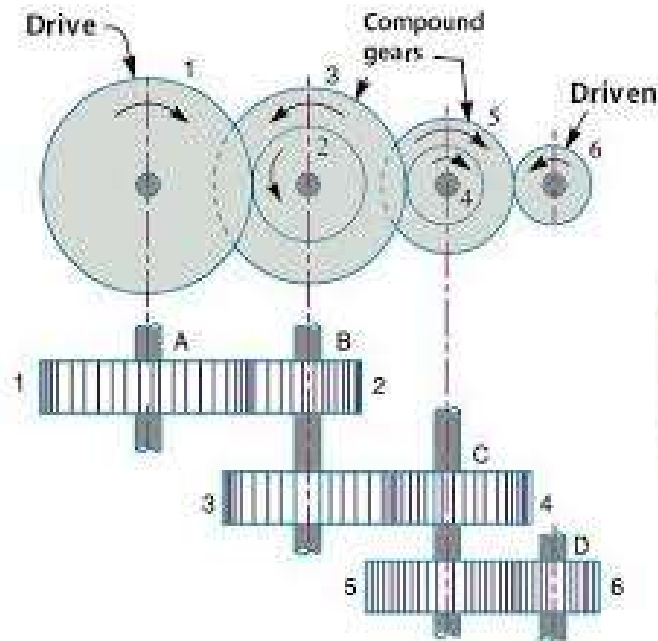


Types of Gear Train

(a) Simple Gear Train



(b) Compound Gear Train and



(c) Epicyclic Gear Train

